input prompt and the final answer (i.e., no intermediate reasoning traces). To enable length control, each prompt x_i is augmented by appending a target length instruction. In particular, we form

$$x_i^{new} = \text{Concat}\Big(x_i, \text{“Think for } n_{gold,i} \text{ tokens.”}\Big),$$

where $n_{gold,i}$ is sampled uniformly from $\mathbb{Z}(n_{min}, n_{max})$. This augmentation yields a new dataset $D^{new} = \{(x_i^{new}, y_{gold,i})\}_{i=1}^{N}$.

We then update LLM_θ using a reinforcement learning objective. In our experiments we adopt GRPO (Shao et al., 2024) (though the method is compatible with other RL algorithms). Our reward function combines two terms: a correctness reward r_c and a length penalty r_{length}. It is defined as

$$r(y, y_{gold}, n_{gold}) = \mathbb{I}(y = y_{gold}) - \alpha \cdot \big|n_{gold} - n_y\big|, \tag{1}$$

where $\mathbb{I}(\cdot)$ is the indicator function, n_y is the generated output length, and α is a scalar that regulates the trade-off between generating the correct answer and meeting the target length. In practice, a lower value of α prioritizes correctness when it is critical, whereas a higher value enforces stricter adherence to the length constraint. Notably, the reward function serves a dual purpose: (a) it encourages the model to produce correct answers while implicitly favoring concise reasoning traces when shorter outputs are requested, and (b) it consistently motivates the model to match the prescribed target length even when a correct answer could be generated with fewer tokens. We refer to the model trained with this objective as L1-Exact.

At inference, the output length is controlled by selecting a fixed target length n_{gold} (or a set of lengths) that is appended uniformly to every test prompt.

Maximum Length Constraint Mode. We further train a variant of L1 called L1-Max, which flexibly generates outputs of varying lengths while respecting a maximum length constraint. This approach is valuable when users prioritize staying within a computational budget rather than adhering to exact generation lengths. To train L1-Max, we fine-tune the L1-Exact model using the same RL framework but with a modified reward function:

$$r(y, y_{gold}, n_{gold}) = \mathbb{I}(y = y_{gold}) \cdot \text{clip}(\alpha \cdot (n_{gold} - n_y) + \delta, 0, 1), \tag{2}$$

where α controls the penalty for length violations. This formulation applies a soft constraint that (1) gradually penalizes outputs exceeding the target length rather than imposing a hard cutoff (which is necessary to ensure gradient propagation in GRPO objective), and (2) incentivizes the model to use fewer tokens when possible without sacrificing correctness. The $\delta = 0.5$ term ensures that correct answers with minor budget violations are still preferred over incorrect answers. Further, L1-Max is trained with dual objective: when the prompt requests an exact length, the model uses Equation 1; otherwise, it defaults to the maximum constraint mode using Equation 2.

4 Experimental Setup

Models and Datasets. We conduct training on the DeepScaleR-Preview-Dataset (Luo et al., 2025), a mathematics dataset consisting of 40K question-answer pairs drawn from AIME, AMC, Omni-Math (Gao et al., 2024) and STILL (Min et al., 2024). We evaluate our models on test sets of 4 different reasoning datasets: AIME 2025, MATH (Hendrycks et al., 2021b), AMC, Olympiad-Bench (He et al., 2024), and additionally GPQA (Rein et al., 2023), LSAT (Zhong et al., 2023), and MMLU (Hendrycks et al., 2021a). Our base model is DeepScaleR-1.5B-Preview, a 1.5B-parameter model originally RL fine-tuned (from DeepSeek-R1-Distill-Qwen-1.5B (DeepSeek-AI et al., 2025)) on this dataset with a 24K token context length. Due to compute constraints, we restrict the maximum context length to 4K tokens during training and to 8K tokens during evaluation. The model is further fine-tuned for 700 steps with LCPO-Exact objective (Equation 1), and the resulting model is referred to as L1-Exact. The model is further RL finetuned for 120 steps with the objective mentioned in Equation 2, and the resulting model is referred to as L1-Max.

Baselines. We evaluate our proposed method against the following baselines:

1.) DeepSeek-R1-Distill-Qwen-1.5B: is the version of Qwen-2.5-1.5B-Instruct finetuned on reasoning traces of DeepSeek's R1 model. For brevity, we refer it as DeepSeek-R1-1.5B.

2.) DeepScaleR-1.5B-Preview: the original model, evaluated without any length control modifications. For brevity, we refer this model as DeepScaleR-24K.

3.) DeepScaleR-1.5B-Preview-4K: a version of DeepScaleR-24K fine-tuned with 4K context length. This is done due to computational constraints of training `LCPO` with long sequence length (such as 24K used in DeepScaleR-24K). The model therefore serves as a fair comparison to `L1`. For brevity, we refer to this model as DeepScaleR-4K.

4.) S1: (Muennighoff et al., 2025) is a budget-forcing method, which controls reasoning length using simple test-time interventions. Concretely, once the maximum token budget is reached, S1 stops further generation and instead inserts "Final Answer" in the prompt to force model to generate the final answer.

Evaluation Protocol. We evaluate our approaches along two dimensions. First, we assess the model's ability to adhere to the targeted length by reporting the mean deviation between the generated token length n_y and the target n_{gold}. Second, we evaluate the overall performance (i.e., problem-solving accuracy) when generating responses at different target lengths. In our experiments, target lengths are selected from $\{512, 1024, 2048, 3600\}$ tokens. For all our experiments, we run 16 random seeds, with a temperature of 0.6.

Hyperparameters and Implementation Details. For GRPO training, we adopt the same hyperparameters as in DeepScaleR-1.5B-Preview. In particular, we use a learning rate of 1e-6 and a batch size of 128. The maximum context length is set to 4K tokens at training time and extended to 8K tokens during evaluation. Training is performed for 700 steps using the VeRL framework (MLSys, 2025). During training, the target length n_{gold} is sampled uniformly from $U(n_{min}, n_{max})$, where we set $n_{min} = 100$ and $n_{max} = 4000$. The balancing parameter α in Equation 1 is fixed at 0.0003. Note that we did not conduct extensive hyperparameter tuning, so one can expect further improvements with additional optimization.

5 Results and Analysis

In this section, we report and analyze the effectiveness of the proposed method (`LCPO`) across various settings and benchmarks. We evaluate our method's relative performance, generalization capability on out-of-domain tasks, controllability of length constraints and competitive performance in short CoT setups, and examine learned reasoning patterns.

`L1` significantly outperforms other length-controlled models while maintaining strong performance. Figure 2 compares performance of `L1`-Exact and `L1`-Max with other baselines across varying generation lengths. Both variants of `L1` achieve superior performance across all token budgets while maintaining strong length control. Compared to S1, the only other method specifically designed for length control, `L1` shows remarkable improvements, over 100-150% relative and 20-25% absolute performance gains at both 512 and 1024 token budgets. This substantial difference can be attributed to two key factors: (1) `L1` intelligently adapts its chain-of-thought to fit within specified length constraints without disrupting the reasoning process, while S1 often truncates mid-reasoning; and (2) `L1` is explicitly trained to generate high-quality reasoning chains of varying lengths, effectively distilling reasoning patterns from longer chains to shorter ones.

Moreover, with `L1`, we observe a log-linear scaling pattern, similar to the prior works O1 and S1 by OpenAI—performance improves linearly with respect to the log-length of generated reasoning chains. However, this scaling curve for `L1` exhibits a notably smaller slope (0.24 vs. 0.37 slope of S1), indicating substantially improved effectiveness at lower token ranges.

`L1`-Exact performs approximately 1% below DeepScaleR-4K, which is the same underlying model as `L1`, but trained without length constraints. However, this difference is primar-

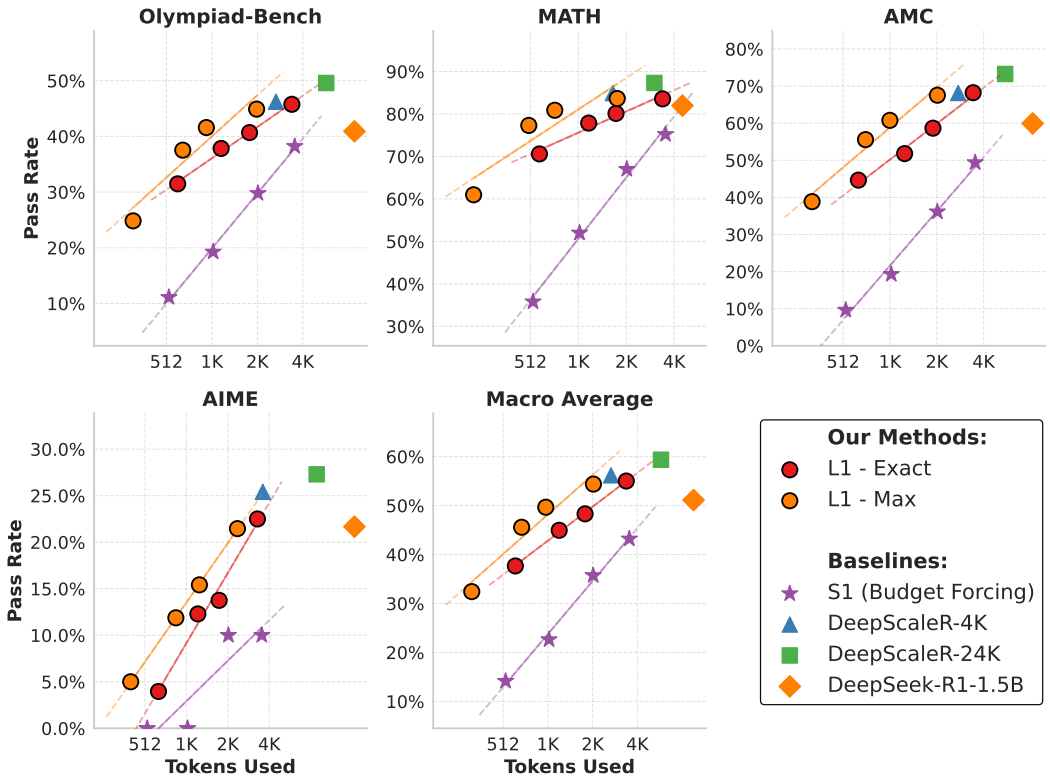

Figure 2: **Performance of L1 under varying token budgets.** Pass rate (y-axis) across benchmarks as a function of tokens used (x-axis). L1-Exact and L1-Max significantly outperform length-controlled baseline (S1) while approaching performance of unconstrained models.

ily observed in the AIME dataset, where unconstrained models can generate very long chains for complex problems. Additionally, L1-Exact allocates the same token budget to all problems regardless of difficulty, potentially using extra tokens on simpler problems. L1-Max effectively alleviates this challenge, matching the performance of DeepScaleR-4K by optimizing token usage based on problem difficulty while respecting the upper ceiling. In doing so, it outperforms even L1-Exact often by up to 2x fewer tokens. L1-Max is particularly valuable when exact token counts are less desirable than a worst-case compute budget. Finally, the scaling trends suggest that with longer context training, L1 would match or even surpass DeepScaleR-24K's performance while maintaining a high degree of length control.

L1 generalizes effectively to out-of-domain (OOD) tasks. We evaluate L1's ability to generalize length control capabilities to domains outside its RL training distribution. We categorize out-of-domain (OOD) datasets: general reasoning datasets GPQA and LSAT that were not explicitly used in L1's training but is likely within DeepSeek-R1-1.5B's training domain and MMLU, which likely falls even outside DeepSeek-R1-1.5B's training distribution.

Figure 3 confirms that L1 generalizes robustly to new domains: performance consistently scales positively with token budget for OOD general reasoning datasets, approaching or matching DeepScaleR-4K benchmarks despite explicit length control constraints. For GPQA and LSAT, we observe the same linear performance scaling trend as in our primary datasets, with L1 matching DeepScaleR-4K's performance at comparable token budgets. This generalization is particularly impressive given that L1 was not explicitly trained on these tasks. For MMLU, we see a less pronounced linear scaling relationship ($R^2 = 0.66$), likely because these knowledge-focused questions benefit less from extended reasoning.